Table of Contents

1. Project Overview

This report presents the development of a predictive system for Universal Plus Bank to identify customers likely to apply for business credit. Anticipating customer behavior enables engagement with key clients and supports retention of high-value customers.

The system was built based on a dataset of 100,000 anonymized customer records, each containing 200 numerical features. The research follows the CRISP-DM workflow, each section detailing actions taken and their rationale, demonstrating how the predictive system delivers actionable insights and aligns with the bank's strategic objectives.

2. Literature Review

Relevant academic papers on predicting business credit applications were reviewed to guide the project, focusing on challenges such as class imbalance, high-dimensional data, model selection, and evaluation of metrics.

Class imbalance is a key challenge in credit datasets, where only a small fraction of customers apply for credit (He & García, 2009). To address this, Anis and Ali (2017) demonstrated that SMOTE generates synthetic minority-class instances to improve detection of rare events, while Ruchay et al. (2023) highlighted that standard accuracy can be misleading in imbalanced datasets. Together, these studies informed the project's decision to balance the dataset using SMOTE combined with random undersampling and prioritize metrics such as Recall, Precision, F1-score, and IBA for evaluation.

Model selection was guided by complementary insights. He et al. (2014) and Mallik et al. (2024) recommended SVM with feature selection and hyperparameter tuning, which influenced the approach to handle high-dimensional data. Ensemble methods were supported by Aburbeian and Ashqar (2023), showing that Random Forest mitigates overfitting while managing imbalanced datasets, providing a strong baseline model. Decision Trees also benefit from careful tuning of depth and split criteria (Mantovani et al., 2023), complementing the ensemble approach by offering interpretability. XGBoost's balanced boosting works better with difficult, large credit data (Carmona, Climent and Momparler, 2019), leading to its selection of the final model. Logistic Regression was retained as a simple benchmark, ensuring proper validation and assessment as emphasized by Dey et al. (2025).

These insights guided the workflow, enabling accurate identification of potential credit applicants, and supporting the bank's strategy.

3. Business Understanding (Phase 1)

Universal Plus Bank aims to proactively identify customers who are likely to apply for business credit. Since only a small share of customers submit applications, the bank requires a predictive model that can highlight meaningful patterns in the data and reliably separate genuine prospects from the wider portfolio. This helps the team focus on customers who show stronger signs of future applications while avoiding outreach that adds little value. A well-designed model will support timely engagement, tailored product offers, and more efficient resource planning. The insights generated will guide prioritization, strengthen relationships, and help the bank manage its future credit pipeline.

4. Data Understanding (Phase 2)

An exploratory data analysis was conducted to inspect outliers, data distribution, missing values, and summary statistics of all features in the dataset, which consisted of 100,000 customer records, 200 variables, an ID code, and a binary target value. According to the box plots and histograms displayed in Figures 1 and 2, there were no outliers, and all features followed a normal distribution. A total of 71 columns contained missing values, of which 68 included 1-5 missing records (<0.005%) and were categorized as the small-missing-value group; var_76 and var_96 had around 5,000 missing records each a (5%), forming the medium-missing-value group; and var_45 contained around 40,000 missing records (40%), representing the high-missing-value group. Furthermore, the dataset exhibited class imbalanced, with the proportion of class 0 to class 1 at approximately 90:10.

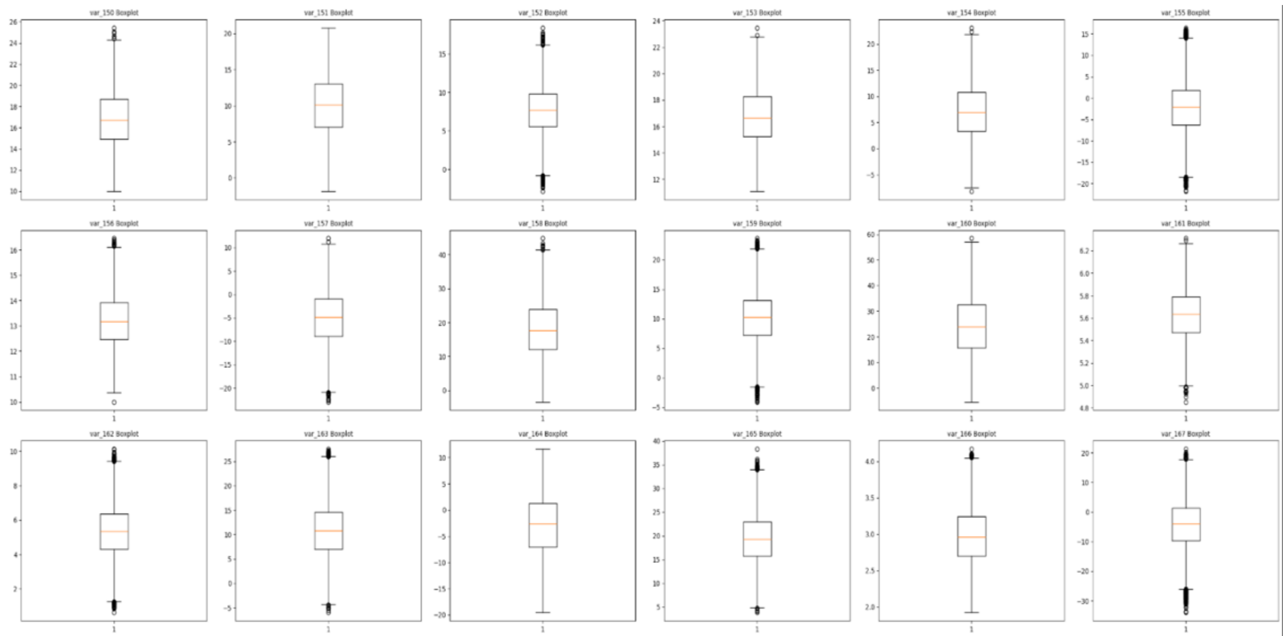

Figure 1: Box-plot distribution of features in the dataset

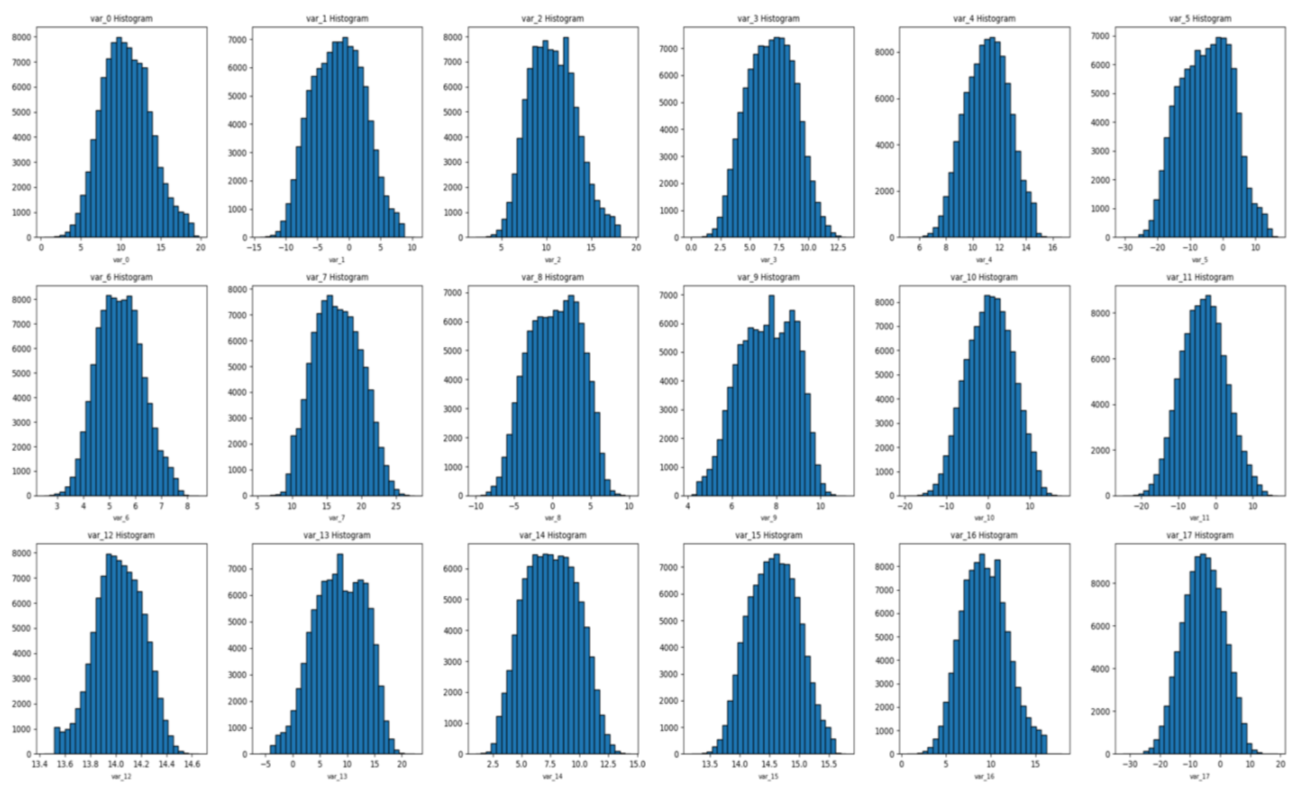

Figure 2: Example of the normal distribution of features in the dataset

5. Data Preparation (Phase 3)

After the data understanding phase, handling missing data was executed as the first step. In general, missing values can be filled by mean, median, or removed from the dataset; however, in this case, imputing missing values could have altered the data distribution, particularly within the high-missing-value group. Conte et al. (2025) suggested that features with more than 40% missing values might likely to distort results; thus, this feature was excluded alternatively. To address missing data less than 5% of records, Lan et al. (2025) proposed that eliminating these records would ensure limited bias and preserve robustness. Therefore, all three groups of missing data were removed, reducing the dataset by nearly 5%, which is acceptable.

Another approach adopted in data preparation was the exclusion of redundant variables such as the ID code, since this feature does not present valuable insights and might cause inaccurate sequences for the model.

Following data cleaning, data partitioning was implemented by distinguishing the target feature from the predictor features. The dataset was segmented to a training set (70%) and a test set (30%), with the test set remaining unmanipulated to ensure impartial model assessment. This proportion of partitioning provided sufficient capacity of both the training data for modelling and the test data for model evaluation. The stratify parameter was applied in both subsets to maintain the equivalent ratio of the original data.

Thereafter, data balancing methodologies, including Random Undersampling and SMOTE, were applied to the training set. Abu Elsoud, E.T. et al. (2024) indicated that Random Undersampling, which randomly eliminated some of the dominant class, performed effectively under high imbalance, and SMOTE is effective because synthetic minority-class cases are not obtained by recreating the data records but generating between existing samples (Anis and Ali, 2017).

6. Modeling (Phase 4)

In the modeling phase, models were developed to predict whether customers were likely to apply for business credit. Several improvements to models were made based on literature review, which included several informative research.

GridSearchCV was used for hyperparameter tuning, which helped improve model performance and reduce overfitting. Through this approach, multiple sets of hyperparameters were used, and the combination that provided the best possible result in terms of performance metrics such as F1, Recall, and Precision was selected.

Interpretability is important for understanding how different features influence the outcome. In this case, the Information Gain was calculated for every feature. This metric measures how much each feature contributes to the final prediction or outcome.

According to the highly imbalanced dataset and its high dimensionality, machine learning algorithms suitable for binary prediction tasks were chosen such as:

1. Decision Trees:

Decision tree is a machine learning model in which decisions are made by splitting data into branches based on feature values with each branch leading to an outcome.

It was implemented primarily as a foundational baseline model. It is highly interpretable and transparent, which makes it easier to explain the prediction methodology and logic, and it is suitable for classification tasks.

In terms of hyperparameters, the model used entropy as the criterion for selecting splits and a maximum depth of 9 to prevent overfitting.

2. Random Forest:

Random forest is an ensemble machine learning model which makes predictions through multiple decision trees on subsets of data.

Random forest models usually outperform single decision trees and perform well in classification tasks as they can capture complex feature interactions.

The model was trained with 100 trees, with the maximum depth set to 10 to control complexity. The maximum number of features per split was set to 4, and the minimum number of samples per leaf was set to 30.

3. Support Vector Machines (SVM):

Support Vector Machines (SVM) are supervised machine learning algorithms that separate two classes using a hyperplane and perform well in binary classification tasks.

SVM was selected due to its strong performance on complex and non-linear problems. Compared with tree-based models, SVM could provide better generalization when classes are difficult to separate.

The model used the RBF kernel and C as 0.1 to prevent overfitting, and gamma was set as 0.01. To support threshold optimization, probability was set as True.

4. XGBoost:

XGBoost is a supervised machine learning algorithm that builds an ensemble of decision trees sequentially, where each tree corrects the errors of the previous ones.

XGBoost normally provides strong predictive performance in real-world applications. In this case, where the minority class was difficult to capture, XGBoost improved recall and F1 score through progressively correcting misclassifications.

The model used 700 estimators with a learning rate of 0.25, a maximum depth of 1 prevented overfitting, a sub sample was set to 0.58 and the fraction of the features for each tree is 0.8.

5. Logistic Regression:

Logistic regression is a machine learning algorithm that predicts the probability of a binary outcome by using a weighted combination of input features.

Logistic regression was included as a benchmark model to assess how a simple approach would perform against more complex algorithms. It is highly interpretable, works well with high-dimensional data, and is suitable for binary tasks such as predicting credit applicants.

For hyperparameters, the maximum number of iterations was set to 30,000, with a fixed random state for reproducibility.

6. LightGBM:

LightGBM is a gradient boosting machine learning algorithm that constructs decision trees in a leaf-wise manner rather than level-wise, which allows it to learn complex patterns efficiently.

This model establishes strong performance in predictive modelling, particularly in tasks involving large and imbalanced datasets. Its speed and scalability allow a large number of trees to be trained quickly while attaining competitive F1 and recall values.

The model was trained with 500 estimators, a learning rate of 0.045, a maximum depth of 15 and 17 leaves. Class weight was set to balance to account for class imbalance.

7. Model Evaluation (Phase 5)

7.1 ROC Curve Analysis

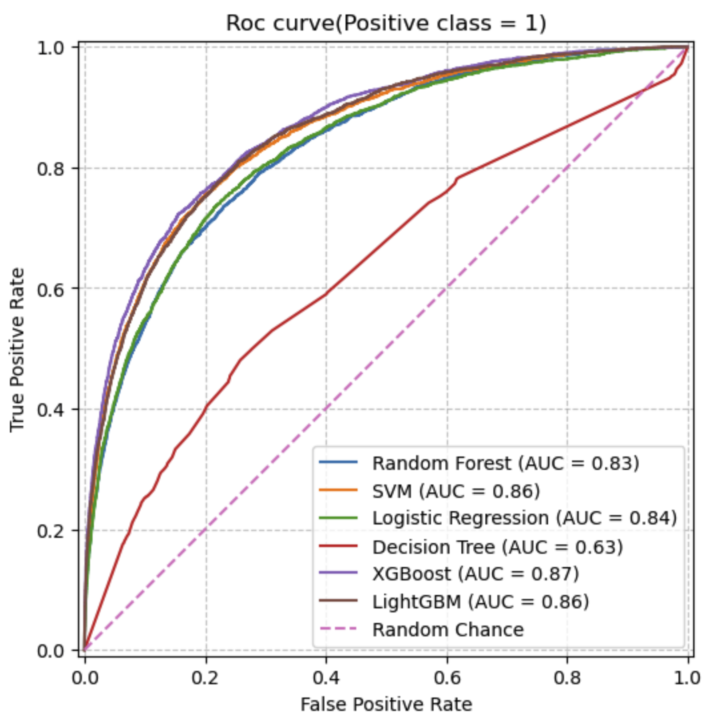

Figure 3: ROC Curve analysis for all the models

7.2 Cumulative Gain Curve Analysis

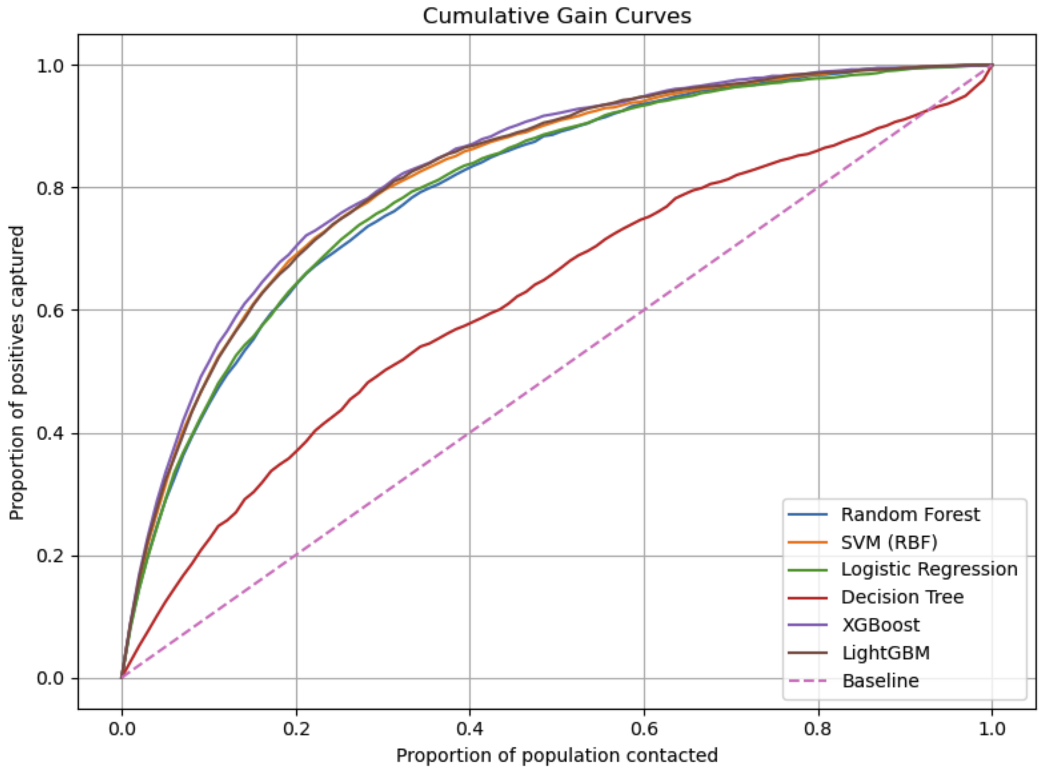

Figure 4 :Cumulative Gain Curve analysis for all the models

Table 1: Rate of Positives Captures for Top 30% in the Cumulative Gain Curve

Model	% Positives Captured (Top 30%)
Decision Tree	~50.21%
Random Forest	~75.39%
Logistic Regression	~76.48%
SVM	~79.71%
LightGBM	~79.85%
XGBoost	~80.27%

7.3 Precision–Recall Curve Analysis

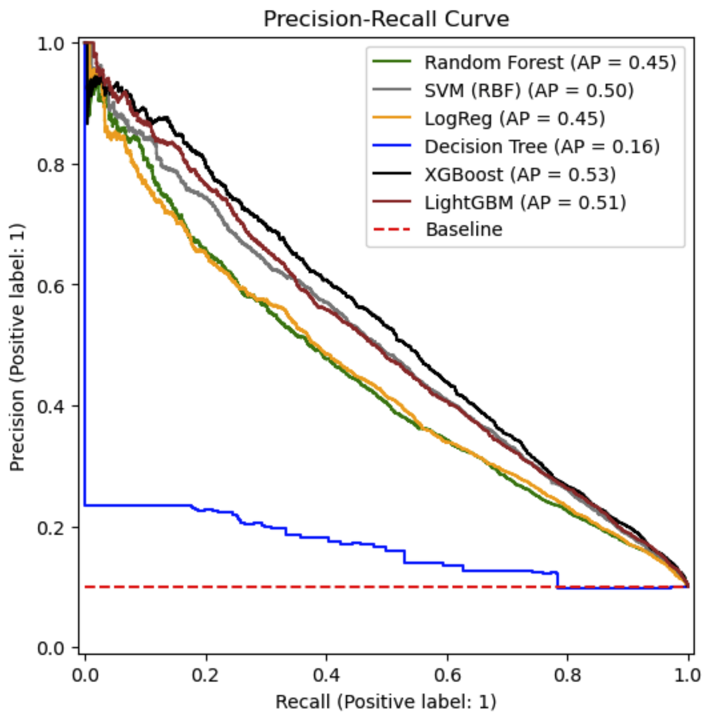

Figure 5: Precision Recall Curve analysis for all the models

7.4 Confusion Matrix Figures

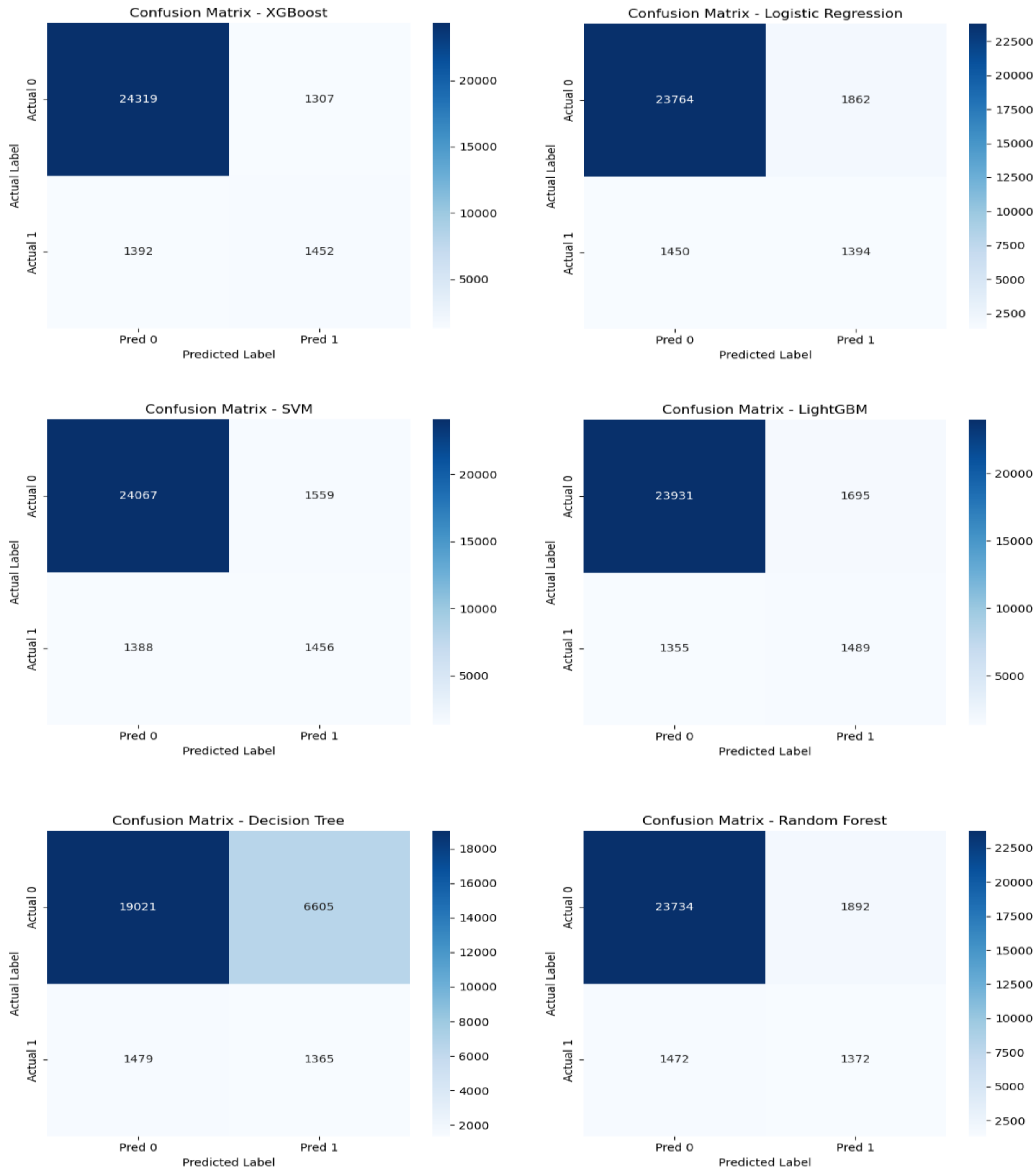

Figure 6: Confusion Matrix assessment for all the models

Table 2: The Performance Metrics for all the models

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.883667	0.428133	0.490155	0.457049
SVM	0.896488	0.482919	0.511955	0.497013
Random Forest	0.881841	0.420343	0.482419	0.449247
Decision Tree	0.716052	0.171267	0.479958	0.252451
XGBoost	<u>*0.905198*</u>	<u>*0.526278*</u>	0.510549	<u>*0.518294*</u>
LightGBM	0.892870	0.467651	<u>*0.523558*</u>	0.494028

* Italics, and underlined for top performer in every metric

The metrics in the model including accuracy, precision, recall, and F1 score were used to evaluate model performance. Due to the imbalanced dataset, the threshold was modified to maximize the F1 score instead of using default threshold, in order to balance precision and recall.

In imbalanced datasets, a PR curve is typically informative than ROC curve. ROC curves can display overly optimistic scenarios in imbalanced datasets because they integrate true negative rates, which are artificially inflated by the dominant negative class, which can potentially mask poor minority class detection.

The PR curve focuses exclusively on positive class performance, which provides a more accurate assessment in imbalanced datasets. The cumulative gain chart was used to assess each model's ability to capture positives when targeting a subset of the population, with XGBoost capturing approximately 80% of positives within the top 30% of predictions.

According to Table 2, XGBoost achieved the highest score across most of the performance metrics evaluated. The confusion matrices displayed in Figure 6 also ensured that XGBoost correctly identifies 1,452 true positives while minimizing false negatives to 1,392, therefore demonstrating a strong accuracy, precision, and recall.

Furthermore, XGBoost maintained relatively low false positives (1,307), significantly outperforming decision trees (6,605), and random forests (1,892). Although LighGBM achieved a good recall, it generated more false positives and negatives than XGBoost. LightGBM reported the highest recall amongst all the models; however, XGBoost outperformed it across the remaining metrics and obtained higher accuracy. Ultimately, the confusion matrix outcomes and the evaluation metrics ensure that XGBoost was the optimal option.

8. Results and Insights

The predictive system developed in this report is well aligned with Universal Plus Bank's goal of proactive customer retention. Prioritizing the F1-score as a balanced metric, the XGBoost classifier emerged as the preferred model, achieving an F1-score of 0.5183 and a recall of 0.5105 on the test set. Although performance on the minority class is constrained by severe class imbalance and the limited usefulness of many anonymous features, these results indicate that, given these data limitations, combining random undersampling, hyperparameter tuning and decision-threshold optimisation produces a robust and practically valuable model.

Table 1 shows that by targeting only the top 30% of customers ranked by the model, the bank can capture more than 80% of future credit applicants, demonstrating strong operational efficiency. This gives Universal Plus a competitive advantage by shifting customer engagement from reactive to proactive. Powered by an XGBoost model within a complete CRISP-DM workflow, the solution is robust and efficient, ensuring resources are precisely allocated to maximise retention of promising customers and strengthen business relationships.

9. AI Declaration

This is to certify that the work we are submitting is our own. All external references and sources are clearly acknowledged and identified within the contents. We are aware of the University of Warwick regulations concerning plagiarism and collusion.

No substantial part of this work has been submitted by us in other assessments for accredited courses of study, and we understand that any such repetition would result in an appropriate reduction in marks.

Generative AI was used in this project to support the quality of our coding, assist our research process, and help ensure that our written explanations were clear and correctly structured. It was used to review academic papers, check coding syntax, explore examples of machine learning techniques, and verify grammar and sentence structure. The tools used were ChatGPT and Gemini. AI assistance was also applied during the coding stage, specifically when implementing the search for the optimal decision threshold and when identifying potential hyperparameters for model tuning.

Reference

Abu Elsoud, E.T., Hassan, M., Alidmat, O., Al Henawi, E., Alshdaifat, N., Igtait, M., Ghaben, A., Katrawi, A. and Dmour, M. (2024) ‘Under sampling techniques for handling unbalanced data with various imbalance rates: a comparative study’, International Journal of Advanced Computer Science and Applications, 15(8), pp.1–10. doi: 10.14569/IJACSA.2024.01508124.

Aburbeian, A.M. and Ashqar, H.I. (2023) ‘Credit Card Fraud Detection Using Enhanced Random Forest Classifier for Imbalanced Data’, arXiv preprint. Available at: https://arxiv.org/abs/2303.06514 (Accessed: 28 November 2025).

Anis, M. and Ali, M. (2017) ‘Investigating the Performance of SMOTE for Class Imbalanced Learning: A Case Study of Credit Scoring Datasets’, European Scientific Journal, 13(33), p.340. doi: 10.19044/esj.2017.v13n33p340.

Carmona, P., Climent, F. and Momparler, A. (2019) ‘Predicting failure in the US banking sector: An extreme gradient boosting approach’, International Review of Economics & Finance, 61, pp.304–323.

Conte, S.L., Fruscoloni, P., et al. (2025) ‘Development and Validation of a NEWS2-Enhanced Multivariable Prediction Model for Clinical Deterioration and In-Hospital Mortality in Hospitalized Adults’, PMC. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC12471953/ (Accessed: 3 December 2025).

Dey, D., Haque, M.S., Islam, M.M., Aishi, U.I., Shammy, S.S., Mayen, M.S.A., Noor, S.T.A. and Uddin, M. (2025) ‘The proper application of logistic regression model in complex survey data: a systematic review’, BMC Medical Research Methodology, 25, Article 15. doi: 10.1186/s12874-024-02454-5.

He, B., Shi, Y., Wan, Q. and Zhao, X. (2014) ‘Prediction of Customer Attrition of Commercial Banks based on SVM Model’, Procedia Computer Science, 31, pp.423–430. Available at: https://doi.org/10.1016/j.procs.2014.05.286 (Accessed: 26 November 2025).

He, H. and García, E.A. (2009) ‘Learning from Imbalanced Data’, IEEE Transactions on Knowledge and Data Engineering, 21(9), pp.1263–1284. doi: 10.1109/TKDE.2008.239.

Lan, X., Mai, H., et al. (2025) ‘Predictive value of combined inflammatory markers for septic cardiomyopathy: a retrospective study’, BMC Infectious Diseases, 25, article no.11447. doi: 10.1186/s12879-025-11447-8.

Mallik, R., Puri, S., Bhansali, A., Manikandan, R. and Rahunathan, L. (2024) ‘Defaulter detection based on Spearman correlation with hyper tuned SVM classification for preventing non-performing assets occurring in bank’, Multimedia Tools and Applications, 84, pp.33071–33093. Available at: https://doi.org/10.1007/s11042-024-20494-3 (Accessed: 26 November 2025).

Mantovani, R.G., Horváth, T., Rossi, A.L.D., Cerri, R., Barbon Junior, S., Vanschoren, J. and de Carvalho, A.C.P.L.F. (2023) ‘Better Trees: An empirical study on hyperparameter tuning of classification decision tree induction algorithms’, arXiv preprint arXiv:1812.02207.